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PROBE HEAD WITH ADJUSTABLE PROTRUSION LENGTH OF PROBE

Abstract

Disclosed is a probe head configured such that the height of a spacer constituted by a plurality of blocks is adjusted, whereby the protrusion length of a probe is adjusted. Since the protrusion length of the probe under a lower plate is adjusted, the number of tests is increased, and therefore the lifespan of the probe head is extended. In addition, work time for replacement and reinstallation of the probe head is shortened, whereby delay of a test process is prevented and process cost is reduced.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a probe head for testing semiconductor elements, and more particularly to a probe head configured such that the height of a spacer constituted by a plurality of blocks is adjusted, whereby the protrusion length of a probe is adjusted.

BACKGROUND ART

[0002] In general, a semiconductor device manufacturing process includes a patterning process of manufacturing semiconductor elements, an electrical die sorting (EDS) process of electrically testing the semiconductor elements to determine whether the semiconductor elements are defective, and an assembly process of integrating the semiconductor elements on a wafer.

[0003] In the EDS process, which is a process of supplying test current to the semiconductor elements and testing electrical signals output from the semiconductor elements to determine whether the semiconductor elements are defective, a probe apparatus configured such that a probe is electrically brought into contact with the semiconductor elements to test performance of the semiconductor elements is widely used.

[0004] The probe apparatus includes a tester configured to supply test current and to test and analyze a signal based thereon, a probe card configured to electrically connect an object to be tested (a semiconductor element) and the tester to each other, and a probe configured to be brought into direct contact with the object to be tested and a printed circuit board of the probe card.

[0005] The probe is generally configured to have a probe head structure in which a plurality of probes is received and assembled such that the probes stably contact the object to be tested and the probe card while maintaining an appropriate contact pressure and durability of the probes is guaranteed even after a plurality of tests.

[0006] In general, the probe head is constituted by a probe and a plate assembly in which the probe is received and assembled. The probe is received in the plate assembly such that a first contact tip and a second contact tip of the probe protrude outwards from a first surface and a second surface of the plate assembly, respectively, so as to electrically contact the printed circuit board and a contact terminal of a semiconductor element to be tested at an appropriate pressure.

[0007] The plate assembly may be configured to have various structures capable of receiving and supporting the probe. For a general vertical probe, the plate assembly mainly includes an upper plate having a receiving hole configured to receive the probe formed therein, the upper plate being configured to support an upper side of the probe, a lower plate having a receiving hole configured to receive the probe formed therein, the lower plate being configured to support a lower side of the probe, and a spacer disposed between the upper plate and the lower plate, the spacer being configured to separate the upper plate and the lower plate from each other by a predetermined distance in order to stably support the probe and to provide a space in which the probe is deformed.

[0008] During testing, a contact tip of the probe of the probe head is worn due to contact pressure thereof or due to scrubbing between the contact tip of the probe and a contact terminal of the object to be tested.

[0009] As a result, the overall lifespan of the probe head is shortened, whereby the number of tests is limited. In addition, work time for replacement and reinstallation of the probe head is added, whereby a test process is delayed and production cost is increased.

DISCLOSURE

Technical Problem

[0010] It is an object of the present invention to provide a probe head configured such that the height of a spacer constituted by a plurality of blocks is adjusted, whereby the protrusion length of a probe is adjusted.

Technical Solution

[0011] In accordance with the present invention, the above and other objects can be accomplished by the provision of a probe head for testing semiconductor elements, the probe head including an upper plate having a first receiving hole formed therein, a lower plate formed spaced apart from the upper plate, the lower plate having a second receiving hole formed therein, a probe coupled to the upper plate and the lower plate such that an upper part of the probe is received in the first receiving hole, whereby an upper tip protrudes upwards from the upper plate, and a lower part of the probe is received in the second receiving hole, whereby a lower tip protrudes downwards from the lower plate, and a spacer formed between the upper plate and the lower plate to separate the upper plate and the lower plate from each other, thereby providing a space portion capable of receiving a middle part of the probe, wherein the spacer is constituted by a plurality of blocks stacked in an upward-downward direction, and the blocks are configured to be selectively removable to adjust the height of the space portion, whereby the protrusion length of the lower tip of the probe under the lower plate is adjustable.

[0012] The spacer may include an upper block, a lower block, and n middle blocks (n being a natural number including 0) provided between the upper block and the lower block.

[0013] The upper block, the lower block, and the middle blocks may be identical in size or shape to each other, the upper block, the lower block, and the middle blocks may be different in size or shape from each other, or at least one of the upper block, the lower block, and the middle blocks may be different in size or shape from the other blocks.

[0014] In addition, the upper block, the lower block, and the middle blocks may be formed so as to have sequentially changed widths or heights in the upward-downward direction, or the upper block, the lower block, and the middle blocks may have different colors.

[0015] In addition, an indicator for position discrimination is provided at a surface of each of the upper block, the lower block, and the middle blocks.

[0016] Any one of a concave-convex structure, a screw engagement structure, and an adhesion structure may be formed at coupling surfaces of each of the upper block and the lower block and a corresponding one of the upper plate and the lower plate that face each other such that the coupling surfaces are coupled to each other.

[0017] A concave-convex structure may be formed at coupling surfaces of the upper block, the lower block, and the middle blocks that face each other such that the coupling surfaces are coupled to each other by concave-convex coupling, a screw engagement structure may be formed at coupling surfaces of the upper block, the lower block, and the middle blocks that face each other such that the coupling surfaces are coupled to each other by screw engagement, or an adhesion structure may be formed at coupling surfaces of the upper block, the lower block, and the middle blocks that face each other such that the coupling surfaces are coupled to each other by adhesion.

[0018] In addition, two or more of a concave-convex structure, a screw engagement structure, and an adhesion structure may be mixedly formed at coupling surfaces of the upper block, the lower block, and the middle blocks that face each other such that the coupling surfaces are coupled to each other.

[0019] At least one of the upper block, the lower block, and the middle blocks may be made of an elastic material.

[0020] The plurality of blocks may be stacked such that an insertion recess is formed in a lower part of the uppermost block and a block located under the uppermost block is inserted into the insertion recess, and the height of the block inserted into the insertion recess may be equal to or less than the depth of the insertion recess.

[0021] The direction in which any one of the plurality of blocks is coupled to the other blocks may be changed to adjust the height of the space portion, and a concave-convex structure or a screw engagement structure is formed at a coupling surface of the block having the changed coupling direction that is to face a coupling surface of a corresponding one of the other blocks.

[0022] The upper plate and the upper block or the lower plate and the lower block may be coupled to each other via a variable fastening member.

[0023] The spacer may be formed so as to have a quadrangular frame shape such that the space portion is formed in a closed state, or the spacer may be formed so as to have a plurality of bridges located at symmetrical points such that the space portion is formed in an open state.

Advantageous Effects

[0024] As is apparent from the above description, the present invention, which relates to a probe head configured such that the height of a spacer constituted by a plurality of blocks is adjusted, whereby the protrusion length of a probe is adjusted, has the following effects.

[0025] First, the present invention has an effect in that the protrusion length of the probe under a lower plate is adjusted, whereby the number of tests is increased, and therefore the lifespan of the probe head is extended.

[0026] Second, the present invention has another effect in that work time for replacement and reinstallation of the probe head is shortened, whereby delay of a test process is prevented and process cost is reduced.

[0027] Third, the present invention has a further effect in that the protrusion length of the probe is adjustable depending on the degree of wear of the probe, whereby contact between a printed circuit board and a contact terminal of an object to be tested at an appropriate pressure is achieved, and therefore more precise and accurate tests are possible while the printed circuit board and the contact terminal of the object to be tested are protected.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] The above and other objects, features and other advantages of the present invention will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0029] FIGS. **1** to **8** are schematic views showing various embodiments of a probe head with an adjustable protrusion length of a probe according to the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0030] The present invention relates to a probe head for testing semiconductor elements, and more particularly to a probe head configured such that the height of a spacer constituted by a plurality of blocks is adjusted, whereby the protrusion length of a probe is adjusted.

[0031] The protrusion length of the probe under a lower plate is adjusted, whereby the number of tests is increased, and therefore the lifespan of the probe head is extended. In addition, work time for replacement and reinstallation of the probe head is shortened, whereby delay of a test process is prevented and process cost is reduced.

[0032] Furthermore, the protrusion length of the probe is adjustable depending on the degree of wear of the probe, whereby contact between a printed circuit board and a contact terminal of an object to be tested at an appropriate pressure is achieved, and therefore more precise and accurate tests are possible while the printed circuit board and the contact terminal of the object to be tested are protected.

[0033] Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings. FIGS. **1** to **8** are schematic views showing various embodiments of a probe head with an adjustable protrusion length of a probe according to the present invention.

[0034] As shown, the probe head with the adjustable protrusion length of the probe **300** according to the present invention includes an upper plate **100** having a first receiving hole **110** formed therein, a lower plate **200** formed spaced apart from the upper plate **100**, the lower plate **200** having a second receiving hole **210** formed therein, a probe **300** coupled to the upper plate **100** and the lower plate **200** such that an upper part of the probe is received in the first receiving hole **110**, whereby an upper tip **310** protrudes upwards from the upper plate **100**, and a lower part of the probe is received in the second receiving hole **210**, whereby a lower tip **320** protrudes downwards from the lower plate **200**, and a spacer **400** formed between the upper plate **100** and the lower plate **200** to separate the upper plate **100** and the lower plate **200** from each other, thereby providing a space portion **500** capable of receiving a middle part of the probe **300**, wherein the spacer **400** is constituted by a plurality of blocks stacked in an upward-downward direction, and the blocks are configured to be selectively removable in order to adjust the height of the space portion **500**, whereby the protrusion length of the lower tip **320** of the probe **300** under the lower plate **200** is adjustable.

[0035] The probe head according to the present invention mainly includes an upper plate **100**, a lower plate **200**, a probe **300** coupled to the upper plate **100** and the lower plate **200**, and a spacer configured to separate the upper plate **100** and the lower plate **200** from each other and to support the upper plate **100** and the lower plate **200**.

[0036] In particular, the spacer **400** according to the present invention is constituted by a plurality of blocks stacked in the upward-downward direction, and the blocks are configured to be selectively removable in order to adjust the height of a space portion **500** defined by the upper plate **100**, the lower plate **200**, and the spacer **400**, whereby the protrusion length of the probe **300** is adjustable.

[0037] In general, while a semiconductor element is tested, a contact tip of the probe **300** is worn due to contact pressure against a printed circuit board and an object to be tested or due to scrubbing between the contact tip of the probe **300** and a contact terminal of the object to be tested.

[0038] As a result, the overall lifespan of the probe head is shortened, whereby the number of tests is limited. In addition, work time for replacement and reinstallation of the probe head is added, whereby a test process is delayed and production cost is increased.

[0039] That is, since the protrusion length of the probe **300** is decreased as the number of tests is increased, the plurality of stacked blocks may be selectively removed in order to adjust the protrusion length of the lower tip **320** of the probe **300** under the lower plate **200** such that the probe **300** is brought into contact with the printed circuit board and the contact terminal of the object to be tested at an appropriate pressure, whereby the number of tests is increased in the state in which more precise and accurate tests are possible.

[0040] The probe **300** according to the present invention may have any shape or may be made of any material in order to test conventional semiconductor elements. In addition, each of the upper plate **100** and the lower plate **200** may have any shape as long as it is possible to stably receive the probe **300** and to guide and support a space portion in which the probe **300** is slid, received, and moved. Furthermore, each of the upper plate **100** and the lower plate **200** may be provided in single or plural in order to stably and effectively support the probe **300** depending on the probe **300**.

[0041] Several thousands to several tens of thousands of probes **300** are coupled to the receiving holes of the upper plate **100** and the lower plate **200**, and are repeatedly slid upwards and downwards and bent in the receiving holes to perform tests. In the present invention, a description will be given based on the structure in which one probe **300** is coupled to the upper plate **100** and the lower plate **200** for sake of convenience.

[0042] A probe head according to an embodiment of the present invention includes an upper plate **100** having a first receiving hole **110** formed therein, a lower plate **200** formed spaced apart from the upper plate **100**, the lower plate **200** having a second receiving hole **210** formed therein, and a probe **300** coupled to the upper plate **100** and the lower plate **200** such that an upper part of the probe is received in the first receiving hole **110**, whereby an upper tip **310** protrudes upwards from the upper plate **100**, and a lower part of the probe is received in the second receiving hole **210**, whereby a lower tip **320** protrudes downwards from the lower plate **200**.

[0043] Here, the probe **300** may be made of an elastic metal or a metal composite material that has elasticity, and a needle type pin, which is usually called a cobra pin, may be provided as the probe **300**. The upper tip **310** protrudes upwards from the upper plate **100**, and the lower part of the probe is received in the second receiving hole **210**, whereby the probe **300** is brought into stable contact with the printed circuit board and the contact terminal of the object to be tested while being bent in the space portion **500**.

[0044] The spacer **400** according to the present invention is formed between the upper plate **100** and the lower plate **200** to separate the upper plate **100** and the lower plate **200** from each other, thereby providing a space portion **500** capable of receiving a middle part of the probe **300**.

[0045] The spacer **400** is formed between the upper plate **100** and the lower plate **200** so as to separate the upper plate **100** and the lower plate **200** from each other and to support the upper plate **100** and the lower plate **200**, and provides a space portion **500** in which the probe **300** may be moved or bent.

[0046] The spacer **400** is formed so as to have a quadrangular frame shape such that the space portion **500** is formed in a closed state or is formed so as to have a plurality of bridges located at symmetrical points such that the space portion **500** is formed in an open state.

[0047] That is, the spacer **400** is formed along the circumference of each of the upper plate **100** and the lower plate **200** at corners or adjacent to the corners thereof in response to the shape of each of the upper plate **100** and the lower plate **200** so as to have a quadrangular frame shape, whereby the space portion **500** is formed so as to be surrounded by the upper plate **100**, the lower plate **200**, and the quadrangular-frame-shaped spacer.

[0048] Alternatively, the spacer **400** is formed so as to have bridges located at symmetrical points, e.g. opposite corners or vertices of the upper plate **100** and the lower plate **200**, or adjacent thereto such that the space portion **500** is formed in an open state.

[0049] The spacer **400** according to the present invention is constituted by a plurality of blocks stacked in the upward-downward direction, and the blocks are configured to be selectively removable in order to adjust the height of the space portion **500**, whereby the protrusion length of the lower tip **320** of the probe **300** under the lower plate **200** is adjustable.

[0050] FIGS. **1** to **8** show various embodiments of the present invention, wherein the upper plate **100**, the lower plate **200**, the spacer **400** formed between the upper plate **100** and the lower plate **200**, and the probe **300** coupled to the upper plate **100** and the lower plate **200** are schematically shown. The spacer **400** is constituted by a plurality of blocks stacked in the upward-downward direction to determine heights L1, L2, and L3 of the space portion **500**.

[0051] When the probe **300** is worn and thus the length of the probe **300** is decreased during testing, one or more of the plurality of blocks constituting the spacer **400** is removed by a length corresponding thereto or by a length necessary to maintain appropriate contact pressure against the printed circuit board and the contact terminal of the object to be tested as needed, whereby the height of the space portion **500** is adjusted, and therefore the protrusion length D of the lower tip **320** of the probe **300** under the lower plate **200** is adjusted.

[0052] The spacer **400** according to the present invention includes an upper block **410**, a lower block **430**, and n middle blocks **420** (n being a natural number including 0) provided between the upper block **410** and the lower block **430**. That is, the spacer **400** according to the present invention is constituted by at least two blocks. When the length of the probe **300** is decreased, at least one of the blocks is removed, whereby the height of the space portion **500** is adjusted, and therefore the protrusion length of the lower tip **320** of the probe **300** under the lower plate **200** is adjusted.

[0053] The upper block **410**, the lower block **430**, and the middle blocks **420** may be identical in size or shape to each other, may be different in size or shape from each other, or at least one of the upper block **410**, the lower block **430**, and the middle blocks **420** may be different in size or shape from the other blocks. The reason for this is that, when the blocks constituting the spacer **400** are identical in size or shape to each other or at least one block is formed so as to be different in size or shape from the other blocks, the degree of freedom in adjusting the height of the space portion **500** is further improved. That is, the protrusion length of the probe **300** is most appropriately adjusted depending on the degree of wear of the probe **300**.

[0054] In addition, the upper block **410**, the lower block **430**, and the middle blocks **420** may be formed so as to have sequentially changed widths or heights in the upward-downward direction or to have different colors. Alternatively, an indicator for discrimination may be further formed on the surface of each block.

[0055] As a result, a block to be removed is easily recognized, whereby the height of the space portion **500** is conveniently adjusted. That is, a block having a specific width, height, or color is easily distinguished from the other blocks, and whereby removal of a wrong block is minimized and removal of a correct block is rapidly and easily achieved when adjusting the height of the space portion **500**.

[0056] In addition, an indicator for discrimination in height or position, e.g. an Arabic numeral, a Korean consonant, or a letter, may be marked on the surface of each block such that the indicator can be recognized from outside, whereby a block having a specific height is accurately and rapidly removed.

[0057] Any one of a concave-convex structure **600**, a screw engagement structure **700**, and an adhesion structure **800** is formed at coupling surfaces of each of the upper block **410** and the lower block **430** and a corresponding one of the upper plate **100** and the lower plate **200** that face each other such that the coupling surfaces are coupled to each other.

[0058] That is, each of the upper block **410** and the lower block **430** and a corresponding one of the upper plate **100** and the lower plate **200** are stably coupled to each other via a coupling means, such as the concave-convex structure **600**, the screw engagement structure **700**, or the adhesion structure **800**, and are easily separated from each other when each block is removed.

[0059] In the concave-convex structure **600**, a concave portion and a convex portion are formed at the coupling surfaces that face each other so as to correspond to each other such that the coupling surfaces are coupled to each other by concave-convex coupling. In the screw engagement structure **700**, screw holes for screw fastening are formed in the coupling surfaces that face each other such that the coupling surfaces are coupled to each other by screw engagement. In the adhesion structure **800**, adhesive members are formed on the coupling surfaces that face each other such that the coupling surfaces are coupled to each other by adhesion.

[0060] In addition, the concave-convex structure **600** may be formed at coupling surfaces of the upper block **410**, the lower block **430**, and the middle blocks **420** that face each other such that the coupling surfaces are coupled to each other by concave-convex coupling, the screw engagement structure **700** may be formed at the coupling surfaces of the upper block **410**, the lower block **430**, and the middle blocks **420** that face each other such that the coupling surfaces are coupled to each other by screw engagement, or the adhesion structure **800** may be formed at the coupling surfaces of the upper block **410**, the lower block **430**, and the middle blocks **420** that face each other such that the coupling surfaces are coupled to each other by adhesion.

[0061] Alternatively, two or more of the concave-convex structure **600**, the screw engagement structure **700**, and the adhesion structure **800** are mixedly formed at the coupling surfaces of the upper block **410**, the lower block **430**, and the middle blocks **420** that face each other such that the coupling surfaces are coupled to each other by two or more of concave-convex coupling, screw engagement, and adhesion.

[0062] That is, not only coupling between each plate and a corresponding block but also coupling between the blocks is stably performed by the coupling means. In addition, block removal is rapidly performed by separation between the concave portion and the convex portion, screw disengagement, or separation between the adhesive members, and then recoupling between the blocks is easily performed by the coupling means.

[0063] Meanwhile, at least one of the upper block **410**, the lower block **430**, and the middle blocks **420** may be made of an elastic material that exhibits higher elasticity than the other blocks. As a result, the height of the space portion **500** is elastically changed, whereby the probe **300** is brought into elastic contact with the printed circuit board and the contact terminal of the object to be tested.

[0064] In another embodiment of the present invention, the spacer **400** is constituted by a plurality of blocks stacked in the upward-downward direction, and the blocks are stacked such that an insertion recess **440** is formed in a lower part of an uppermost block and a block located under the uppermost block is inserted into the insertion recess **440**, wherein the height of the block inserted into the insertion recess **440** is equal to or less than the depth of the insertion recess **440**.

[0065] When an upper block **410** is removed, the height of the space portion **500** is set by a middle block **420** (or a lower block **430**) that was inserted into the insertion recess **440** of the upper block **410**, and the protrusion length of the lower tip **320** of the probe **300** is adjusted by the difference in height between the upper block **410** and the middle block **420** (or the lower block **430**).

[0066] In another embodiment of the present invention, the direction in which any one of the plurality of blocks is coupled to the other blocks may be changed to adjust the height of the space portion **500**.

[0067] That is, the direction in which the blocks constituting the spacer **400** are coupled to each other is changed to change the height of the spacer **400**. The direction in which the blocks are stacked horizontally or vertically or the direction in which the blocks are coupled to each other is changed based on the difference in horizontal or vertical height between the blocks, and neighboring blocks are recoupled to each other, whereby the height of the spacer **400** is changed.

[0068] When the direction in which the blocks are coupled to each other is changed, as described above, the concave-convex structure **600** or the screw engagement structure **700** is formed at coupling surfaces of the neighboring blocks that are to face each other. Alternatively, the concave-convex structure **600** or the screw engagement structure **700** is formed at coupling surfaces of one of the blocks stacked in the state in which the coupling direction thereof is changed and the upper plate **100** or the lower plate **200** that are to face each other, whereby coupling therebetween is stably performed.

[0069] In another embodiment of the present invention, the upper plate **100** and the upper block or the lower plate **200** and the lower block may be coupled to each other via a variable fastening member **900**.

[0070] The variable fastening member **900** is constituted by a screw thread having a predetermined length and a screw head. The variable fastening member **900** is configured such that, when the upper plate **100** and the upper block are coupled to each other by screw engagement and when the lower plate **200** and the lower block are coupled to each other by screw engagement, the distance between each of the plates and a corresponding one of the blocks coupled thereto is decreased, and when screw engagement therebetween is released, the distance between each of the plates and a corresponding one of the blocks coupled thereto is increased. Consequently, the height of the space portion **500** is further adjusted through coupling and separation between the plate and the block using the variable fastening member **900**.

[0071] That is, the height of the space portion **500** may be basically adjusted by block removal, and additionally the height of the space portion **500** may be finely adjusted using the variable fastening member **900**.

[0072] Hereinafter, various embodiments the present invention will be described with reference to the accompanying drawings. In the drawings showing the embodiments of the present invention, the height of each block, the protrusion length of the lower tip **320** of the probe **300**, etc. are somewhat exaggerated for convenience of description. Actually, the height or shape of each block is set in consideration of the degree of wear of the probe **300**, and a change in height of the blocks due to at least block removal or block direction change is set so as to be similar to the degree of wear of the probe **300**.

First Embodiment

[0073] FIG. **1** shows a first embodiment of the present invention, wherein an upper plate **100**, a lower plate **200**, a spacer **400** formed between the upper plate **100** and the lower plate **200**, and a probe **300** coupled to the upper plate **100** and the lower plate **200** are schematically shown.

[0074] The spacer **400** is constituted by three blocks stacked in the upward-downward direction, whereby a space portion **500** has a height L1. That is, the spacer **400** is constituted by three blocks, i.e. an upper block **410**, a middle block **420**, and a lower block **430**, whereby the space portion **500** has a height L1, and a lower tip **320** of the probe **300** under the lower plate has a protrusion length D.

[0075] In the first embodiment of the present invention, the upper block **410**, the middle block **420**, and the lower block **430** constituting the spacer **400** are identical in shape. When the length of the probe **300** is decreased, one of the blocks is removed to adjust the protrusion length of the probe **300**.

[0076] That is, when the probe **300** is worn during testing, whereby the protrusion length of the lower tip **320** of the probe **300** under the lower plate is decreased, scrubbing between the lower tip and a contact terminal of an object to be tested is not sufficiently performed or appropriate contact pressure against a printed circuit board and the contact terminal of the object to be tested is not maintained. In this case, one of the blocks constituting the spacer **400** is removed to reduce the height of the space portion **500** (L1−>L2, L1>L2), whereby the protrusion length of the lower tip **320** of the probe **300** is readjusted to D.

Second Embodiment

[0077] FIG. **2** shows a second embodiment of the present invention, wherein an upper plate **100**, a lower plate **200**, a spacer **400** formed between the upper plate **100** and the lower plate **200**, and a probe **300** coupled to the upper plate **100** and the lower plate **200** are schematically shown.

[0078] The second embodiment of the present invention is identical to the first embodiment of the present invention except that the blocks have different shapes. That is, the widths of the blocks are sequentially changed in the upward-downward direction, whereby it is possible to easily recognize at least one block to be removed.

Third Embodiment

[0079] FIG. **3** shows a third embodiment of the present invention, wherein an upper plate **100**, a lower plate **200**, a spacer **400** formed between the upper plate **100** and the lower plate **200**, and a probe **300** coupled to the upper plate **100** and the lower plate **200** are schematically shown.

[0080] The third embodiment of the present invention is identical to the first embodiment of the present invention except that the direction in which at least one of the blocks and a corresponding one of the other blocks are coupled to each other is changed and adjacent blocks are recoupled to each other to adjust the height of a space portion **500**.

[0081] In the third embodiment of the present invention, the direction of an upper block **410** is changed and the upper block **410** is recoupled to a middle block **420**, whereby the height of the space portion **500** is reduced from L1 to L2, and the direction of a lower block **430** is changed and the lower block **430** is recoupled to the middle block **420**, whereby the height of the space portion **500** is reduced from L2 to L3. As a result, the protrusion length of the probe **300** is adjusted to D.

Fourth Embodiment

[0082] FIG. **4** shows a fourth embodiment of the present invention, wherein an upper plate **100**, a lower plate **200**, a spacer **400** formed between the upper plate **100** and the lower plate **200**, and a probe **300** coupled to the upper plate **100** and the lower plate **200** are schematically shown.

[0083] In the fourth embodiment of the present invention, the spacer **400** is constituted by a plurality of blocks stacked in the upward-downward direction, wherein a middle block **420** is inserted into an insertion recess **440** of an upper block **410**, and a lower block **430** is inserted into an insertion recess **440** of the middle block **420**. The height of each block inserted into the insertion recess **440** is equal to or less than the depth of the insertion recess **440**.

[0084] When the length of the probe **300** is decreased, the upper block **410** is removed. As a result, the height of a space portion **500** is set (L1−>L2) by the middle block **420** that was inserted into the insertion recess **440** of the upper block **410**, and the protrusion length of a lower tip **320** of the probe **300** is adjusted by the difference in height between the upper block **410** and the middle block **420**.

Fifth Embodiment

[0085] FIG. **5** shows a fifth embodiment of the present invention, wherein an upper plate **100**, a lower plate **200**, a spacer **400** formed between the upper plate **100** and the lower plate **200**, and a probe **300** coupled to the upper plate **100** and the lower plate **200** are schematically shown.

[0086] As shown in FIG. **5**, the upper plate **100** and an upper block are coupled to each other via a variable fastening member **900**, and the lower plate **200** and a lower block are coupled to each other via another variable fastening member **900**.

[0087] The variable fastening member **900** is constituted by a screw thread having a predetermined length and a screw head. The variable fastening member **900** is configured such that, when the upper plate **100** and the upper block are coupled to each other by screw engagement and when the lower plate **200** and the lower block are coupled to each other by screw engagement, the distance between each of the plates and a corresponding one of the blocks coupled thereto is decreased, and when screw engagement therebetween is released, the distance between each of the plates and a coupled thereto is corresponding one of the blocks increased. Consequently, the height of the space portion **500** is further adjusted (L1−>L2) through coupling and separation between the plate and the block using the variable fastening member **900**.

[0088] That is, the height of the space portion **500** may be basically adjusted by block removal, and additionally the height of the space portion **500** may be finely adjusted using the variable fastening member **900**, whereby the protrusion length of the probe **300** is finely adjusted.

Sixth Embodiment

[0089] FIGS. **6** to **8** show a coupling means configured to couple the blocks constituting the spacer **400** according to the present invention to each other and a coupling means configured to stably couple the upper plate **100** and the upper block **410** to each other and to stably couple the lower plate **100** and the lower block **410** to each other.

[0090] The coupling means may be basically applied to the first to fifth embodiments.

[0091] FIG. **6** shows the state in which the blocks are coupled to each other through a concave-convex structure **600**, FIG. **7** shows the state in which the blocks are coupled to each other through a screw engagement structure **700**, and FIG. **8** shows the state in which the blocks are coupled to each other through an adhesion structure **800** and each of blocks and a corresponding one of the plates are coupled to each other via the screw engagement structure **700**.

[0092] As described above, the present invention relates to a probe head for testing semiconductor elements, and more particularly to a probe head configured such that the height of a spacer constituted by a plurality of blocks is adjusted, whereby the protrusion length of a probe is adjusted.

[0093] The protrusion length of the probe under a lower plate is adjusted, whereby the number of tests is increased, and therefore the lifespan of the probe head is extended. In addition, work time for replacement and reinstallation of the probe head is shortened, whereby delay of a test process is prevented and process cost is reduced.

[0094] Furthermore, the protrusion length of the probe is adjustable depending on the degree of wear of the probe, whereby contact between a printed circuit board and a contact terminal of an object to be tested at an appropriate pressure is achieved, and therefore more precise and accurate tests are possible while the printed circuit board and the contact terminal of the object to be tested are protected.

Claims

1. A probe head for testing semiconductor elements, the probe head comprising: an upper plate having a first receiving hole formed therein; a lower plate formed spaced apart from the upper plate, the lower plate having a second receiving hole formed therein; a probe coupled to the upper plate and the lower plate such that an upper part of the probe is received in the first receiving hole, whereby an upper tip protrudes upwards from the upper plate, and a lower part of the probe is received in the second receiving hole, whereby a lower tip protrudes downwards from the lower plate; and a spacer formed between the upper plate and the lower plate to separate the upper plate and the lower plate from each other, thereby providing a space portion capable of receiving a middle part of the probe, wherein the spacer is constituted by a plurality of blocks stacked in an upward-downward direction, and the blocks are configured to be selectively removable to adjust a height of the space portion, whereby a protrusion length of the lower tip of the probe under the lower plate is adjustable.
2. The probe head according to claim 1, wherein the spacer comprises: an upper block; a lower block; and n middle blocks (n being a natural number including 0) provided between the upper block and the lower block.
3. The probe head according to claim 2, wherein the upper block, the lower block, and the middle blocks are identical in size or shape to each other, the upper block, the lower block, and the middle blocks are different in size or shape from each other, or at least one of the upper block, the lower block, and the middle blocks is different in size or shape from the other blocks.
4. The probe head according to claim 2, wherein the upper block, the lower block, and the middle blocks are formed so as to have sequentially changed widths or heights in the upward-downward direction, or the upper block, the lower block, and the middle blocks have different colors.
5. The probe head according to claim 2, wherein an indicator for position discrimination is provided at a surface of each of the upper block, the lower block, and the middle blocks.
6. The probe head according to claim 2, wherein any one of a concave-convex structure, a screw engagement structure, and an adhesion structure is formed at coupling surfaces of each of the upper block and the lower block and a corresponding one of the upper plate and the lower plate that face each other such that the coupling surfaces are coupled to each other.
7. The probe head according to claim 2, wherein a concave-convex structure is formed at coupling surfaces of the upper block, the lower block, and the middle blocks that face each other such that the coupling surfaces are coupled to each other by concave-convex coupling.
8. The probe head according to claim 2, wherein a screw engagement structure is formed at coupling surfaces of the upper block, the lower block, and the middle blocks that face each other such that the coupling surfaces are coupled to each other by screw engagement.
9. The probe head according to claim 2, wherein an adhesion structure is formed at coupling surfaces of the upper block, the lower block, and the middle blocks that face each other such that the coupling surfaces are coupled to each other by adhesion.
10. The probe head according to claim 2, wherein two or more of a concave-convex structure, a screw engagement structure, and an adhesion structure are mixedly formed at coupling surfaces of the upper block, the lower block, and the middle blocks that face each other such that the coupling surfaces are coupled to each other.
11. The probe head according to claim 2, wherein at least one of the upper block, the lower block, and the middle blocks is made of an elastic material.
12. The probe head according to claim 2, wherein the plurality of blocks is stacked such that an insertion recess is formed in a lower part of an uppermost block and a block located under the

uppermost block is inserted into the insertion recess, and a height of the block inserted into the insertion recess is equal to or less than a depth of the insertion recess.

13. The probe head according to claim 1, wherein a direction in which any one of the plurality of blocks is coupled to the other blocks is changed to adjust the height of the space portion.

14. The probe head according to claim 13, wherein a concave-convex structure or a screw engagement structure is formed at a coupling surface of the block having the changed coupling direction that is to face a coupling surface of a corresponding one of the other blocks.

15. The probe head according to claim 2, wherein the upper plate and the upper block or the lower plate and the lower block are coupled to each other via a variable fastening member.

16. The probe head according to claim 1, wherein the spacer is formed so as to have a quadrangular frame shape such that the space portion is formed in a closed state, or the spacer is formed so as to have a plurality of bridges located at symmetrical points such that the space portion is formed in an open state.